

Prime factorization

What are the prime factors of $N = 299$?

Step 1: find $x(\text{mod } N) \not\equiv \pm 1$ but $x^2(\text{mod } N) \equiv 1$

$$x^2 = 1 + kN$$

$$x^2 - 1 = kN$$

$$(x-1)(x+1) = k \times N = k \times p \times q$$

For $x = -1, \dots, 300$,

$$\{x, x^2(\text{mod } N)\} = \{\{-1,1\}, \{0,0\}, \{1,1\}, \{2,4\}, \{3,9\}, \{4,16\}, \{5,25\}, \{6,36\}, \{7,49\}, \{8,64\}, \{9,81\}, \{10,100\},$$

$$\{11,121\}, \{12,144\}, \{13,169\}, \{14,196\}, \{15,225\}, \{16,256\}, \{17,289\}, \dots, \{111,62\}, \{112,285\},$$

$$\{113,211\}, \{114,139\}, \{115,69\}, \{116,1\}, \{117,234\}, \{118,170\}, \{119,108\}, \{120,48\}, \{121,289\}, \dots, \{177,233\},$$

$$\{178,289\}, \{179,48\}, \{180,108\}, \{181,170\}, \{182,234\}, \{183,1\}, \{184,69\}, \{185,139\}, \{186,211\}, \dots, \{294,25\},$$

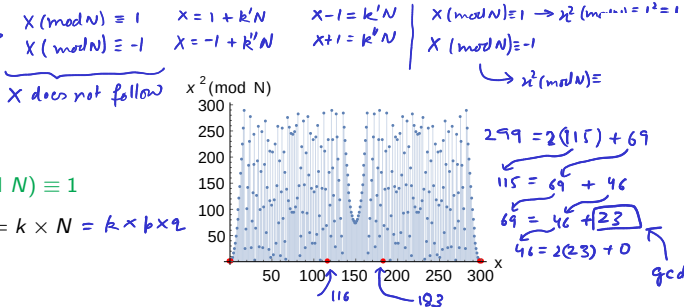
$$\{295,16\}, \{296,9\}, \{297,4\}, \{298,1\}, \{299,0\}, \{300,1\}\}$$

$x = 116$

Step 2: $p = \text{gcd}(x-1, N) = \text{gcd}(115, 299) = 23$
 $q = \text{gcd}(x+1, N) = \text{gcd}(117, 299) = 13$

$$\frac{x-1}{N} = \frac{5}{13}, \quad \frac{x+1}{N} = \frac{9}{23}$$

Euclid's algorithm is an efficient way to compute the greatest common divisor.



$$298 \text{ mod } 299 \equiv -1$$

$$(300)^2 \text{ mod } 299 \equiv 1$$

$$300 \text{ mod } 299 \equiv 1$$

$$(116)^2(\text{mod } 299) \equiv 1$$

$$116(\text{mod } 299) \not\equiv \pm 1$$

video

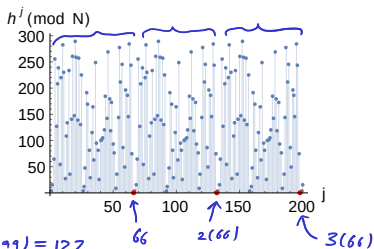
Prime factorization is equivalent to order finding (1)

(p, 2)

Step 1: How to find $x \pmod{N} \not\equiv \pm 1$ but $x^2 \pmod{N} \equiv 1$?

Pick a random integer $h < N$ and compute $h^j \pmod{N}$ for $j = 1, \dots$ till you find **smallest** $j = r$ such that $h^r \pmod{N} \equiv 1$. That r will be the **order**. If r is **even**, then $x = h^{(r/2)}$. The probability of r being even and $\underline{h^{(r/2)} \pmod{N} \not\equiv -1}$ is at least $\frac{1}{2}$.

$$h^{(r/2)} \pmod{N} \not\equiv 1$$



$$1024 \pmod{299} \equiv 127$$

For $h = 4 < 299 = N$ and $j = 1, \dots, 200$,

$\{j, h^j \pmod{N}\} = \{\{1, 4\}, \{2, 16\}, \{3, 64\}, \{4, 256\}, \{5, 127\}, \{6, 209\}, \{7, 238\}, \dots, \{61, 186\}, \{62, 146\}, \{63, 285\}, \{64, 243\}, \{65, 75\}, \{66, 1\}, \{67, 4\}, \{68, 16\}, \{69, 64\}, \{70, 256\}, \dots, \{130, 243\}, \{131, 75\}, \{132, 1\}, \{133, 4\}, \{134, 16\}, \{135, 64\}, \dots, \{195, 285\}, \{196, 243\}, \{197, 75\}, \{198, 1\}, \{199, 4\}, \{200, 16\}\}$

Quantum computer is used to comp $h^j \pmod{N}$

$$4^{66} \pmod{299} \equiv 1$$

$$4^{33} \pmod{299} \not\equiv -1 \checkmark$$

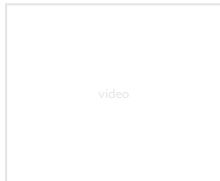
$$r = 66, x = h^{(r/2)} = 4^{33} = 73786976294838206464$$

Step 2: $p = \gcd(x - 1, N) = \gcd(73786976294838206463, 299) = 23$
 $q = \gcd(x + 1, N) = \gcd(73786976294838206465, 299) = 13$

When it does not work: $3^{33} \pmod{299} \equiv 1$, $10^{66} \pmod{299} \equiv 1$ and $10^{33} \pmod{299} \equiv -1 \equiv 298$

When $h = 3$

$h = 10$



Quantum algorithm for order-finding

$$h=4 \quad f(0)=1 \rightarrow |1\rangle \equiv |0000\rangle$$

$$f(1)=4 \rightarrow |4\rangle \equiv |0010\rangle$$

$$f(2)=16 \rightarrow |16\rangle \equiv |1000\rangle$$

Cyclic permutation: $U|f(l)\rangle = |f(l+1)\rangle$ where $f(l+r) = f(l) := h^l \pmod N$ and $l = 0, \dots, r-1$.

$$|0\rangle^{\otimes m} = |0\rangle|0\rangle|0\rangle \dots |0\rangle \equiv |00\dots 0\rangle$$

$$\begin{pmatrix} |u_0\rangle \\ |u_1\rangle \\ \vdots \\ |u_{r-1}\rangle \end{pmatrix} = \frac{1}{\sqrt{r}} \underbrace{\begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & \omega^{-1} & \dots & \omega^{-(r-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-(r-1)} & \dots & \omega^{-(r-1)(r-1)} \end{pmatrix}}_{F_r^{-1}} \begin{pmatrix} |f(0)\rangle \\ |f(1)\rangle \\ \vdots \\ |f(r-1)\rangle \end{pmatrix}$$

$i = \sqrt{-1}$

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{e=0}^{r-1} \omega^{-se} |f(e)\rangle$$

$$U|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{e=0}^{r-1} \omega^{-se} |f(e+1)\rangle$$

$$= \frac{1}{\sqrt{r}} \sum_{e'=0}^{r-1} \omega^{-s(e'-1)} |f(e')\rangle$$

$$= \omega^s \left[\frac{1}{\sqrt{r}} \sum_{e'=0}^{r-1} \omega^{-se'} |f(e')\rangle \right]$$

$e' = e+1$
 $e = e'-1$

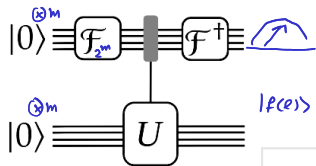
Eigenvalue eqn: $U|u_s\rangle = \omega^s |u_s\rangle$, where $\omega = \exp(i \frac{2\pi}{r})$ is r th root of unity and $s = 0, \dots, r-1$.

$$|0\rangle|0\rangle \xrightarrow{\otimes^m} \left[\frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle |0\rangle \right]^{\otimes m}$$

$$\rightarrow \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle |f(j)\rangle \text{ superposition}$$

$$= \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} \omega^{js} |j\rangle |u_s\rangle$$

$$\rightarrow \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left| \frac{s 2^m}{r} \right\rangle |u_s\rangle$$



$$|0\rangle^{\otimes m} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{2^m \times 1}$$

$$|f(e)\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \omega^{es} |u_s\rangle$$

$$\frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} \exp\left(i \frac{2\pi}{2^m} \left(\frac{2^m}{r} s\right) j\right) |j\rangle$$

$$\left| \frac{s 2^m}{r} \right\rangle |u_s\rangle$$

$\frac{s 2^m}{r} \rightarrow r$ by continued fractions algorithm

In general, $\frac{s 2^m}{r}$ is not an integer.

